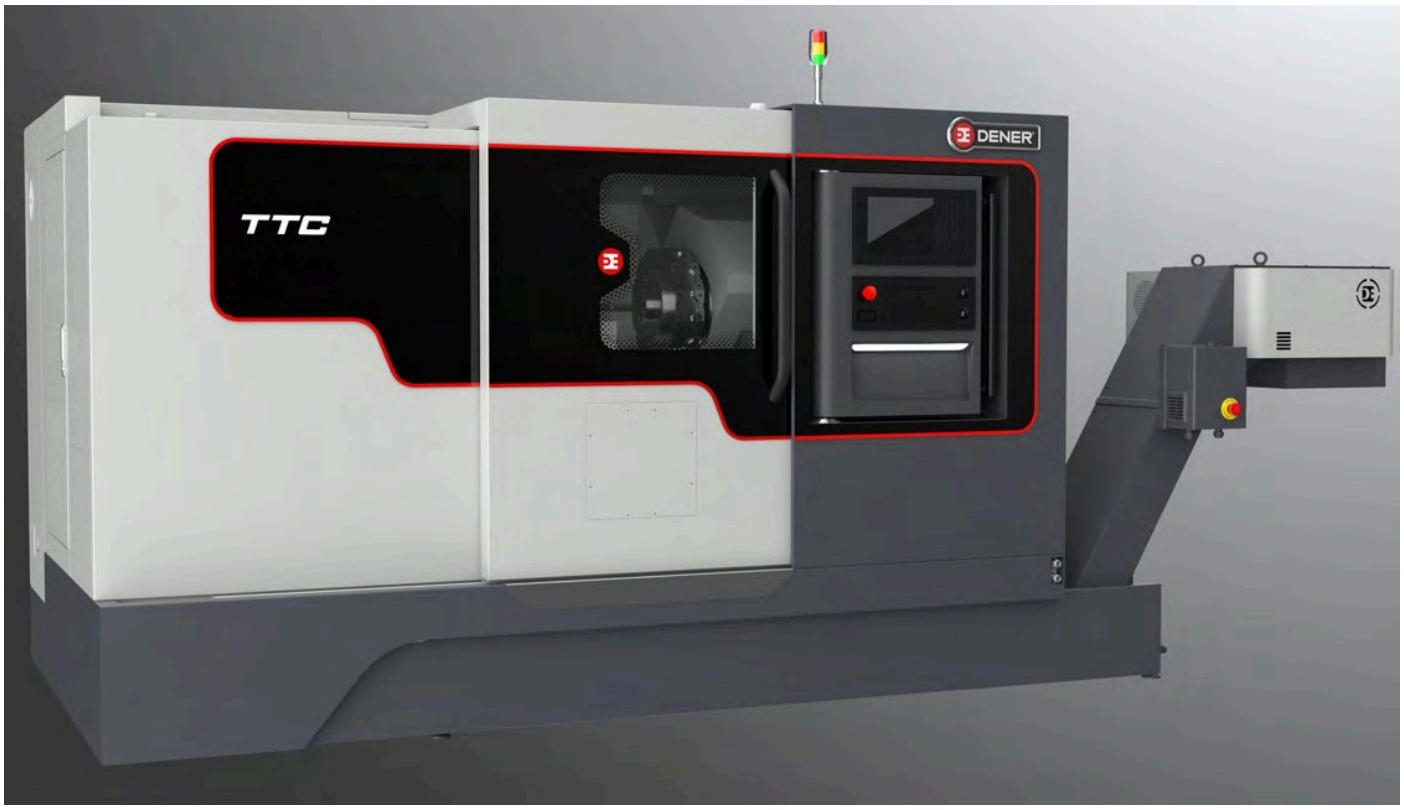
"THE FUTURE IS HERE" · SOLD & SERVICED IN CANADA BY NOVOTECH

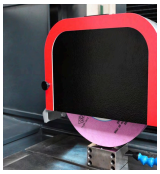
CNC LATHES
TTC · TTH SERIES

TURNING · THREADING · DRILLING · C-AXIS MILLING









Since 1974
MANUFACTURING EXPERIENCE

6 – 12"
HYDRAULIC CHUCK

45°
SLANT BED

Canada
PARTS & TECHNICAL SUPPORT

NovoTech Machine Tools Inc.
1365 Mid-Way Blvd, 26, Mississauga, ON L5T 2J5, Canada
NovoTechMachineTools.com | info@NovoTechMachineTools.com | +1 647 526 5510

MANUFACTURER · ESTABLISHED 1974

ABOUT DENER

Dener Makina was established in 1974 in the Kayseri Old Industrial Zone. Since 2000, the company has focused on the production of CNC Sheet Metal Processing Machines, manufacturing a wide range of products including Guillotine Shears, Hydraulic Press Brakes, Ball Screw Press Brakes, Servo-Electric Press Brakes, Plasma Cutting Machines, and Fiber Laser Cutting Machines.

In 2016, the company made a major investment by expanding its facilities to include 1,670,000 sq ft (155,000 m^2) of factory buildings and 290,000 sq ft (27,000 m^2) of social facilities on a 14,000,000 sq ft (1,300,000 m^2) site within the İncesu Organized Industrial Zone (O.S.B). In this new phase, Dener Makina began the production of CNC Metal Processing Machines such as Taksan Integrated Systems, Double Column Machining Centers, 5-Axis Simultaneous Machining Centers, Vertical Machining Centers, Lathes, and Grinding Machines.

With its extensive production range, over half a century of experience, skilled workforce, and multinational R&D center, Dener Makina holds a leading position globally by providing high-value industrial investment products in the fields of CNC Sheet Metal Processing and CNC Metal Processing Machines.

INCLUDED WITH EVERY NOVOTECH MACHINE

- Made in Turkey with European components
- CSA-approved electrical panel
- 1-year warranty
- **7-day money-back guarantee** (in-stock machines only)
- Spare parts & technical support readily available in Canada
- **Order Today with Just 10% Down — 0% Down Financing Available**
- **Installation and training available**

SOLD & SERVICED IN CANADA

by NovoTech Machine Tools — Mississauga, ON

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DENER MAKINA · İNCESU ORGANIZED INDUSTRIAL ZONE, KAYSERI

DENER PRODUCTION CAMPUS



Dener production campus, Kayseri — 1,670,000 sq ft (155,000 m^2) of factory buildings on a 14,000,000 sq ft (1,300,000 m^2) site

- SHEET PROCESSING MACHINES PRODUCTION PLANT
- MACHINE TOOLS PRODUCTION PLANT
- PRODUCTION PLANT
- SHEET WELDING FACTORY
- SHEET CLADDING FACTORY
- TECHNOLOGY CENTER
- ADMINISTRATIVE BUILDING
- SOCIAL FACILITIES
- MAIN ENTRANCE
- VISITOR PARKING AREA
- HELICOPTER PAD

52+
YEARS OF EXPERIENCE

100+
EXPORT COUNTRIES

14,000,000 sq ft
PRODUCTION CAMPUS (1,300,000 M^2)

TTC-TTH SERIES CNC LATHES

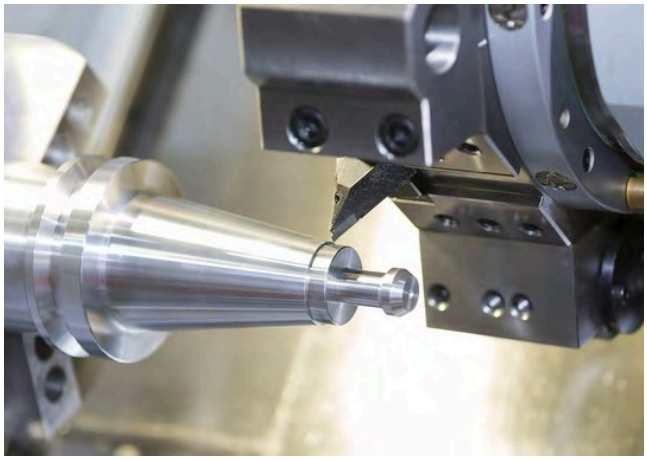
OVERVIEW



TTC CNC lathe

With the CNC lathe, precise internal and external diameter turning, internal and external threading and drilling are carried out. With the C-axis option, milling operations can also be performed.

- Specially manufactured precision, backlash-free double-nut ball screws
- Powerful, fast serial automatic tool-changer turret
- Rigid spindles supported by super-precision bearings
- Stable travel on linear guides supported at many points
- Very precise positioning and repeatability
- Quiet, vibration-free, high cutting capability
- One-piece 45° slant, robust cast body
- Fast service and spare parts support



TTC-TTH SERIES CNC LATHES

STANDARD / OPTIONAL EQUIPMENT

STANDARD EQUIPMENT

- Mitsubishi M80A-8A
- Roller linear guides
- C1 – C3 precision ball screws
- BMT / slot type powerful turret
- Fully enclosed cabin
- Chip conveyor
- Automatic fill-and-drain lubrication system
- Automatic coolant system
- Oil and water separator unit
- 3-phase 380 / 200 V – 50 / 60 Hz external transformer
- Foot pedal
- Front door safety switch
- Hydraulic power unit
- 6" – 8" – 10" – 12" hydraulic chuck, soft and hard jaws
- Tool holders
- In-cabin lighting and alarm / warning lamps
- Electrical cabinet cooling system
- Portable handwheel
- Levelling bolts and plates
- Tailstock
- Air and water gun
- Tool bag & tools
- CE

OPTIONAL EQUIPMENT

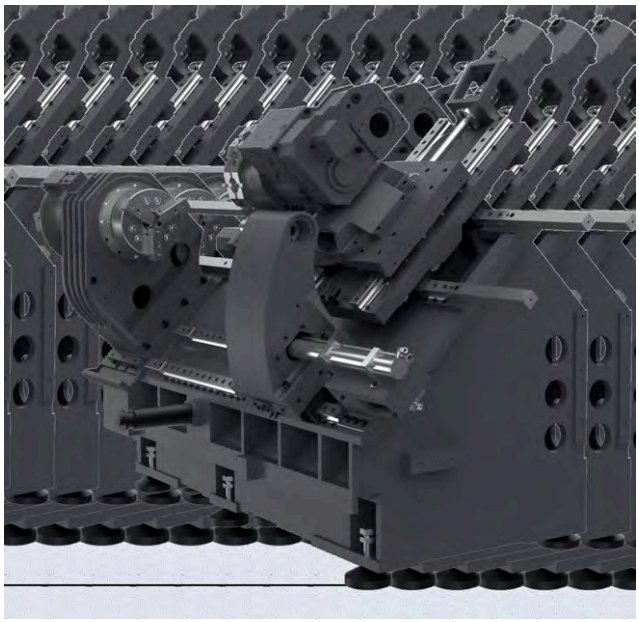
- FANUC 0i TFP
- C axis
- Tool measuring system
- Workpiece measuring system
- Automatic bar feeder unit
- Air conditioner for the electrical cabinet
- Automatic door
- Chip cart
- Mist extraction unit
- Linear encoder
- Programmable tailstock
- Parts catcher





TTC-TTH SERIES CNC LATHE

BODY STRUCTURE AND WORKING AREA



The 45° slant cast body with the turret, spindle and tailstock

BODY STRUCTURE

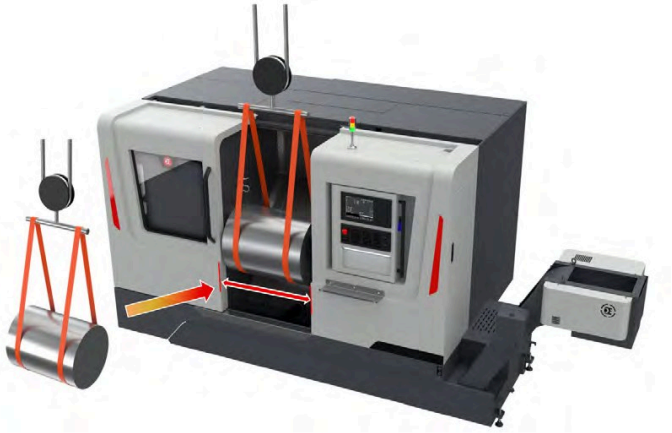
- High speed
- Precision
- Long life
- High durability
- Vibration damping

WORKING AREA

Working area	TTC-6	TTC-8	TTH-8	TTH-10	TTH-12
Travel – X axis	7.5" (190 mm)	7.5" (190 mm)	11.8" (300 mm)	11.8" (300 mm)	11.8" (300 mm)
Travel – Z axis	17.7" (450 mm)	17.7" (450 mm)	29.9" (760 mm)	29.9" (760 mm)	29.9" (760 mm)

PART LOADING

Part loading	TTC-6	TTC-8	TTH-8	TTH-10	TTH-12
Door opening	27.2" (690 mm)	27.2" (690 mm)	29.1" (740 mm)	29.1" (740 mm)	29.3" (745 mm)

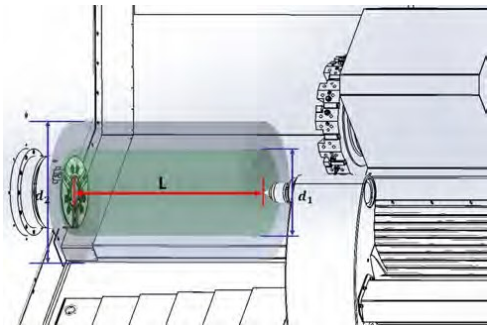


Loading a heavy part by crane through the door

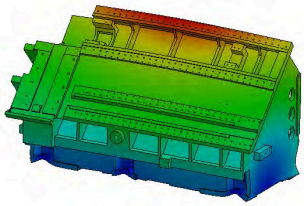
TTC-TTH SERIES CNC LATHE

MACHINING CAPACITY AND STRUCTURE

Machining capacity	TTC-6	TTC-8	TTH-8	TTH-10	TTH-12
Ød – recommended turning diameter	6.7″ (170 mm)	8.3″ (210 mm)	8.3″ (210 mm)	10″ (254 mm)	12″ (304 mm)
$Ød_1$ – max. turning diameter	12″ (305 mm)	12″ (305 mm)	20.8″ (528 mm)	20.8″ (528 mm)	19.2″ (488 mm)
$Ød_2$ – swing over bed	22″ (560 mm)	22″ (560 mm)	27.6″ (700 mm)	27.6″ (700 mm)	27.6″ (700 mm)
L – max. turning length	17″ (433 mm)	16″ (406 mm)	26.8″ (680 mm)	26″ (660 mm)	24.2″ (614 mm)





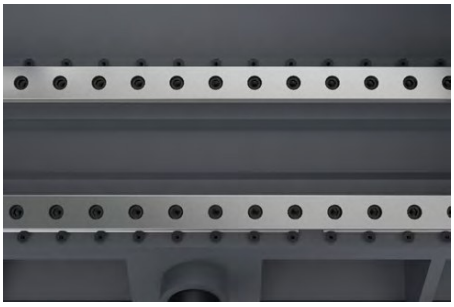


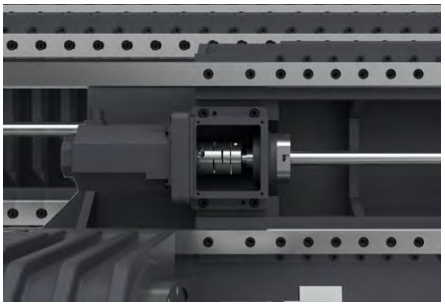
RIGID STRUCTURE

Effective damping of the cutting forces generated during machining, thanks to the massive 45° main body structure.

FINITE ELEMENT METHOD

The optimum body structure obtained using the finite element method.





ULTRA-PRECISE LINEAR EQUIPMENT

Ultra-precise, ground, preloaded and backlash-free ball screws and linear bearing elements for precise positioning, repeatability, minimum friction, rigid support and quiet operation.

DIRECT-DRIVE SERVO MOTORS

Direct connection between the motor and the ball screw for backlash-free, precise motion.

TTC-TTH SERIES CNC LATHES

TURRET, SPINDLE AND TAILSTOCK





TURRET

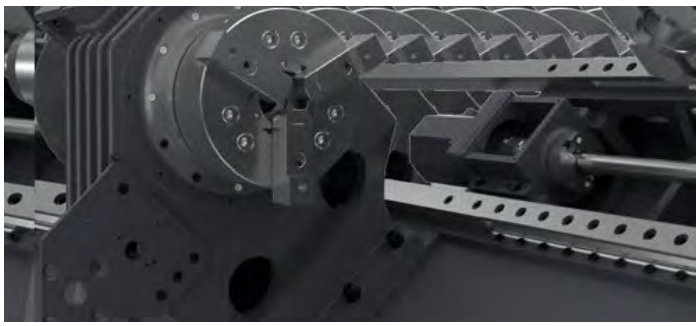
Fast tool change, high machining performance, precise positioning and powerful hydraulic clamping with the 12-station, electro-mechanical, servo-controlled slot type turret.

C-AXIS TURRET (OPTION)

Machining on the C axis, fast tool change, high machining performance, precise positioning and powerful hydraulic clamping with the new-generation BMT type, 12-station (all 12 live tools) electro-mechanical servo turret.

SPINDLE

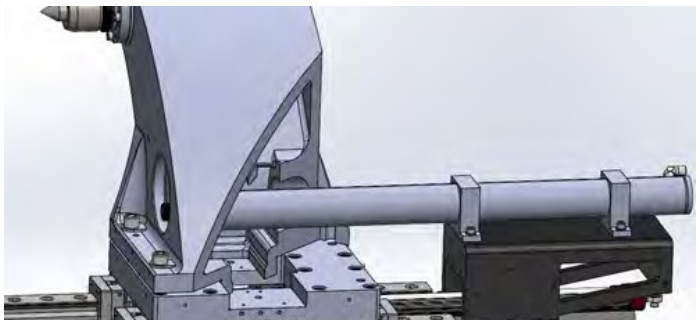
Reinforced with roller bearings, with high heat dissipation, low vibration and quiet operation: high durability and excellent machining.





TAILSTOCK

Easy positioning with hydraulically controlled movement independent of the turret axis, and powerful braking.





Tailstock axis

TTC-TTH SERIES CNC LATHES

CHIP CONVEYOR, COOLANT AND CONTROL UNIT

CHIP CONVEYOR

Trouble-free transport and discharge of the long, coarse and spiral chips of metal, fiber, teflon, delrin and similar materials.

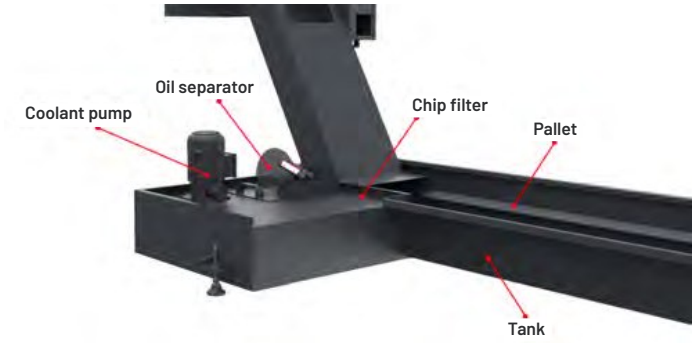






COOLANT TANK

Optimum cooling and easy maintenance with a 37 – 53 US gal (140 – 200 L), self-filtering, functional tank design.



CHIP CART (OPTION)

Easy storage and discharge of all kinds of chips, with its tilting and braking features.



CONTROL UNIT

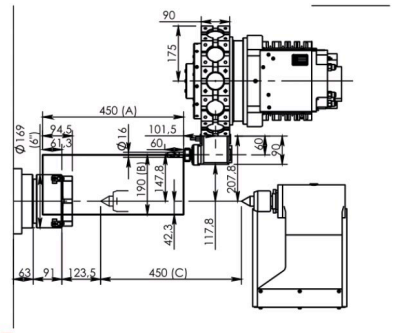
Easy to use, with up-to-date technological modules, an operator-friendly interface and an ergonomic design.



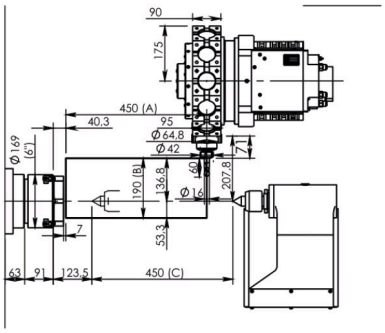


TTC-TTH SERIES · MACHINING DISTANCES

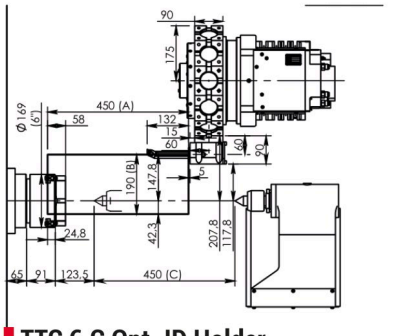
TTC 6 WORKING RANGE



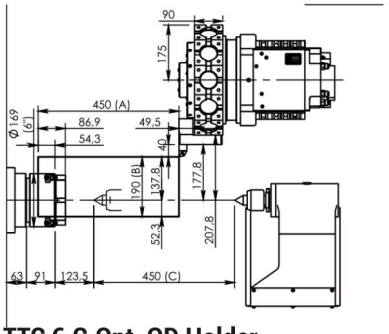
TTC 6 C Opt. Angular Milling Holder



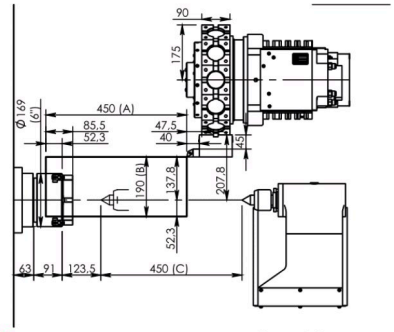
TTC 6 C Opt. Straight Milling Holder



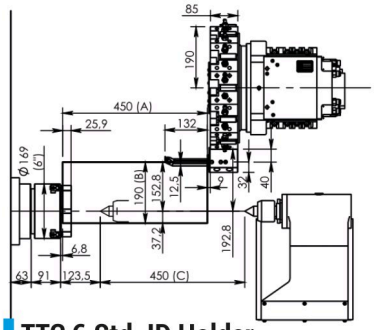
TTC 6 C Opt. ID Holder



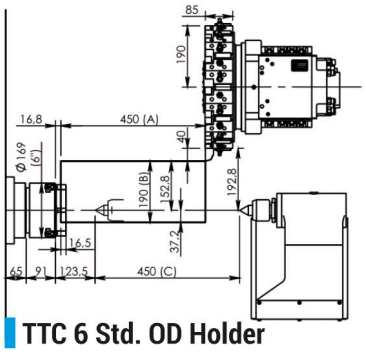
TTC 6 C Opt. OD Holder



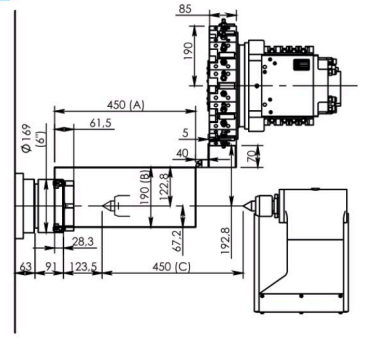
TTC 6 C Opt. Face Tool Holder



TTC 6 Std. ID Holder



TTC 6 Std. OD Holder

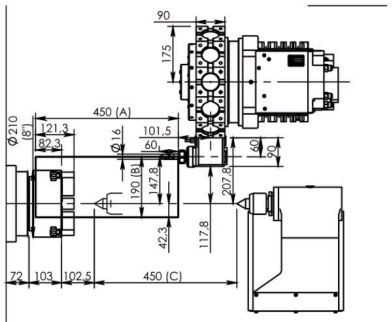


TTC 6 Std. Face Tool Holder

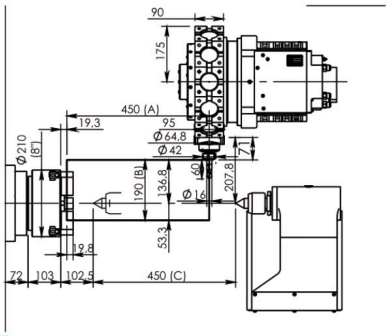
Dimensions on the drawings are in millimetres (1" = 25.4 mm). A: Z-axis travel · B: X-axis travel · C: tailstock travel. C Opt. = with the C-axis turret (option); Std. = standard turret.

TTC-TTH SERIES · MACHINING DISTANCES

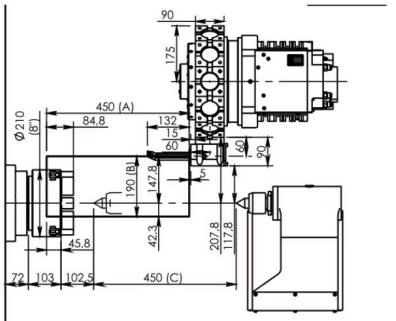
TTC 8 WORKING RANGE



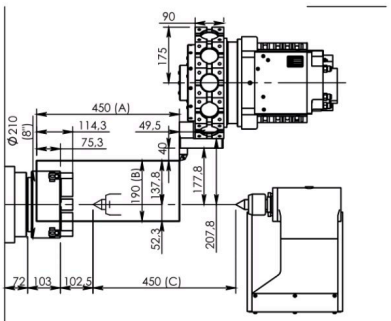
TTC 8 C Opt. Angular Milling Holder



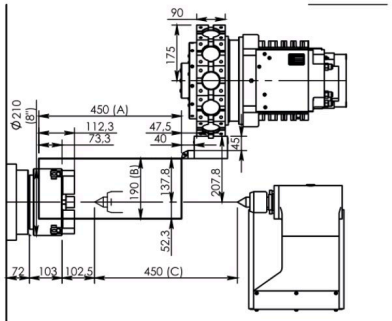
TTC 8 C Opt. Straight Milling Holder



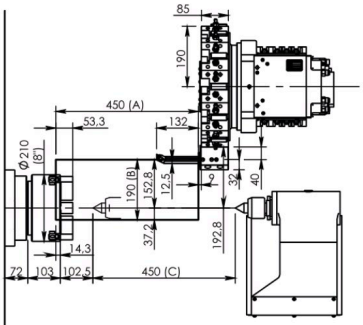
TTC 8 C Opt. ID Holder



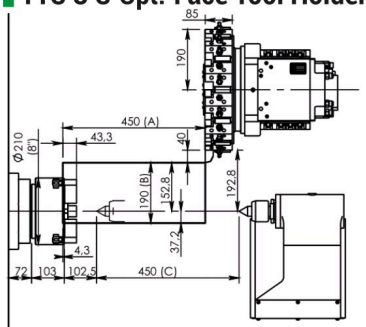
TTC 8 C Opt. OD Holder



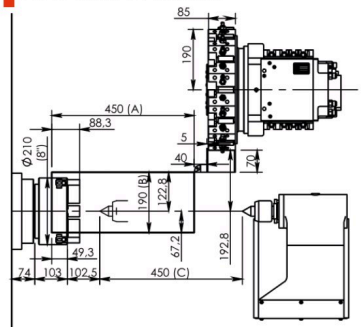
TTC 8 C Opt. Face Tool Holder



TTC 8 Std. ID Holder



TTC 8 Std. OD Holder



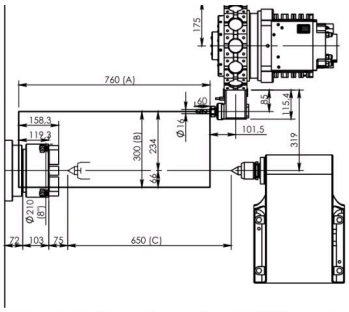
TTC 8 Std. Face Tool Holder

A: Z Axis Travel B:X Axis Travel C: Tailstock Travel

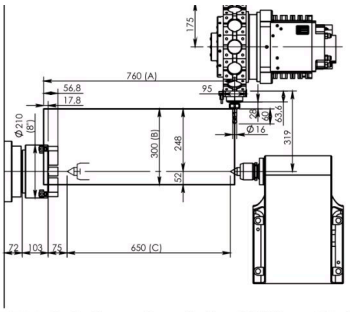
Dimensions on the drawings are in millimetres (1" = 25.4 mm). A: Z-axis travel · B: X-axis travel · C: tailstock travel. C Opt. = with the C-axis turret (option); Std. = standard turret.

TTC-TTH SERIES · MACHINING DISTANCES

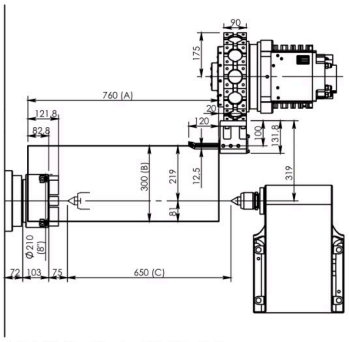
TTH 8 WORKING RANGE



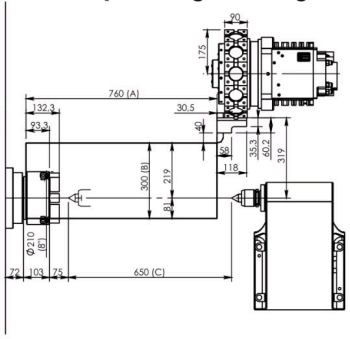
TTH 8 C Opt. Angular Milling Holder



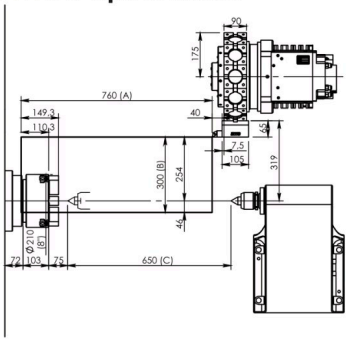
TTH 8 C Opt. Straight Milling Holder



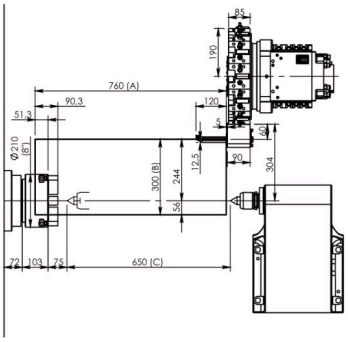
TTH 8 C Opt. ID Holder



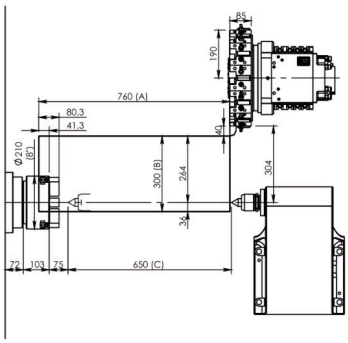
TTH 8 C Opt. OD Holder



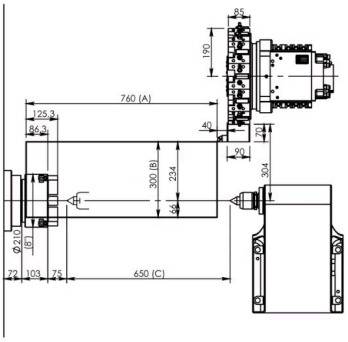
TTH 8 C Opt. Face Tool Holder



TTH 8 Std. ID Holder



TTH 8 Std. OD Holder

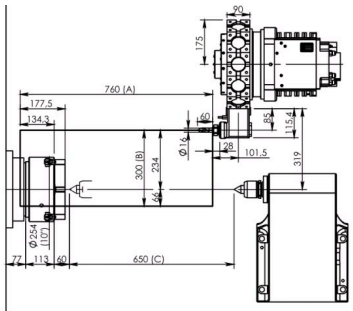


TTH 8 Std. Face Tool Holder

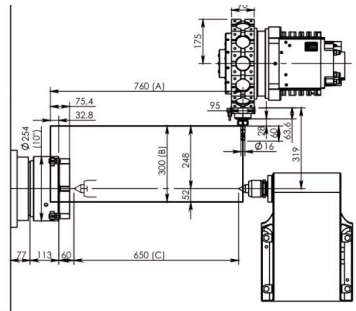
Dimensions on the drawings are in millimetres (1" = 25.4 mm). A: Z-axis travel · B: X-axis travel · C: tailstock travel. C Opt. = with the C-axis turret (option); Std. = standard turret.

TTC-TTH SERIES · MACHINING DISTANCES

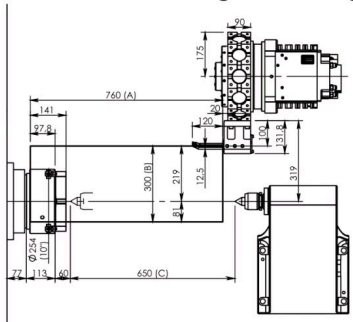
TTH 10 WORKING RANGE



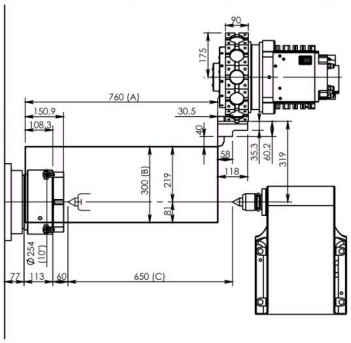
TTH 10 C Opt. Angular Milling Holder



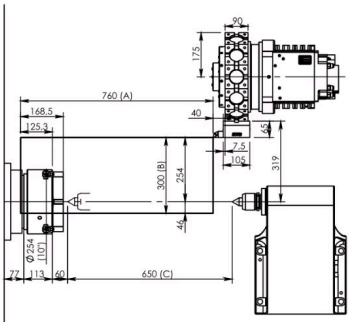
TTH 10 C Opt. Straight Milling Holder



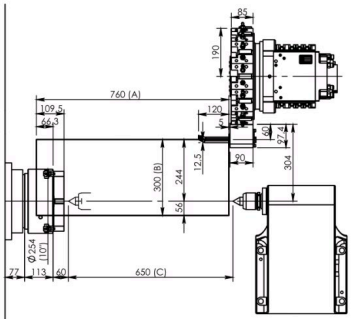
TTH 10 C Opt. ID Holder



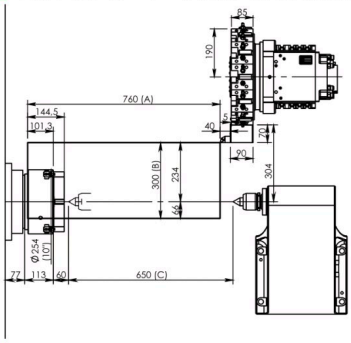
TTH 10 C Opt. OD Holder



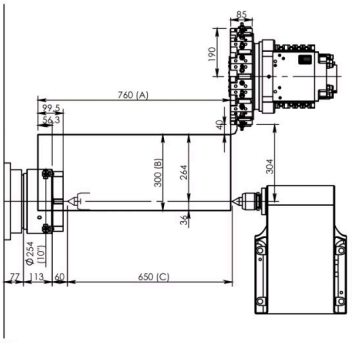
TTH 10 C Opt. Face Tool Holder



TTH 10 Std. ID Holder



TTH 10 Std. Face Tool Holder



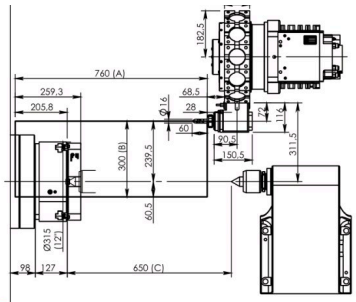
TTH 10 Std. OD Holder

A: Z Axis Travel B:X Axis Travel C: Tailstock Travel

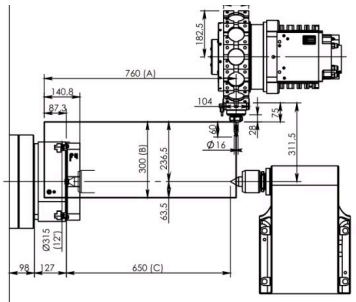
Dimensions on the drawings are in millimetres (1" = 25.4 mm). A: Z-axis travel · B: X-axis travel · C: tailstock travel. C Opt. = with the C-axis turret (option); Std. = standard turret.

TTC-TTH SERIES · MACHINING DISTANCES

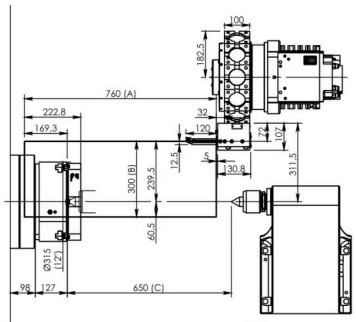
TTH 12 WORKING RANGE



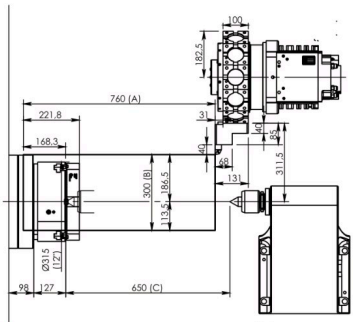
TTH 12 C Opt. Angular Milling Holder



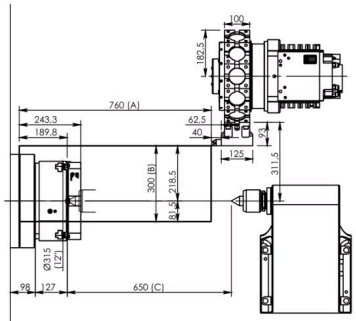
TTH 12 C Opt. Straight Milling Holder



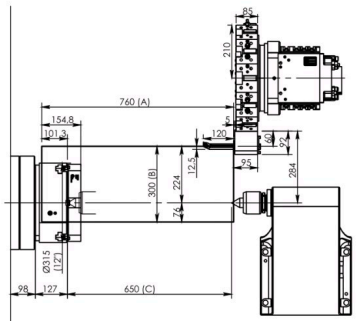
TTH 12 C Opt. ID Holder



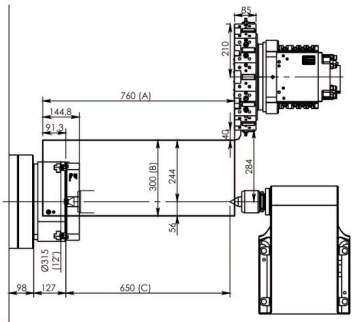
TTH 12 C Opt. OD Holder



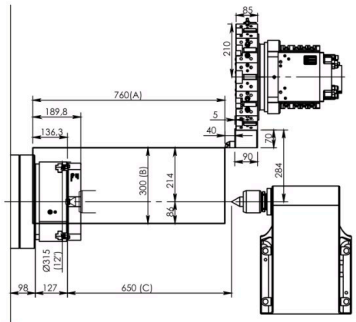
TTH 12 C Opt. Face Tool Holder



TTH 12 Std. ID Holder



TTH 12 Std. OD Holder

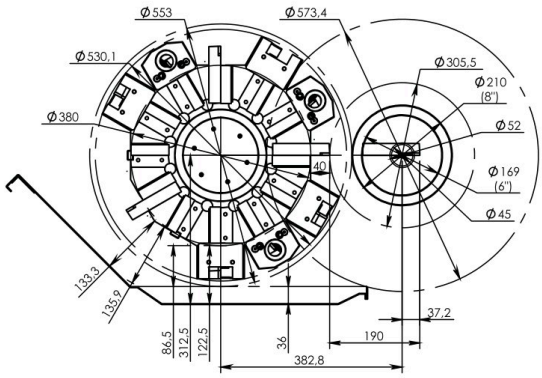


TTH 12 Std. Face Tool Holder

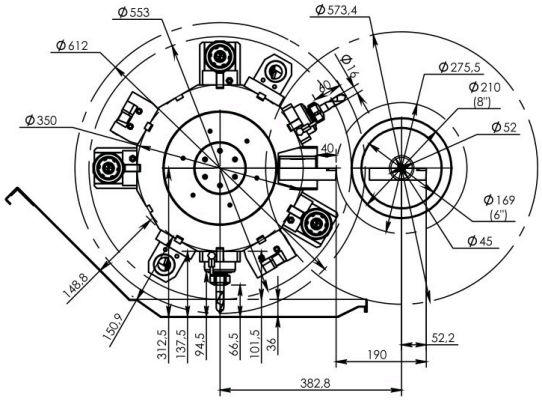
Dimensions on the drawings are in millimetres (1" = 25.4 mm). A: Z-axis travel · B: X-axis travel · C: tailstock travel. C Opt. = with the C-axis turret (option); Std. = standard turret.

TTC-TTH SERIES · TOOL INTERFACE

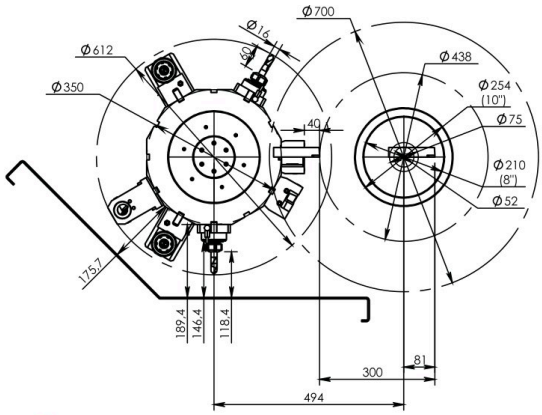
TOOL INTERFERENCE



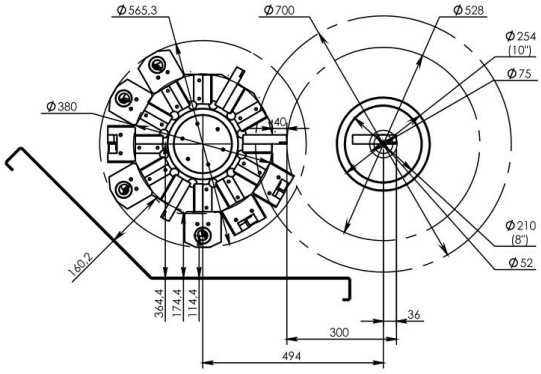
TTC 6-8 Std. Tool Interference



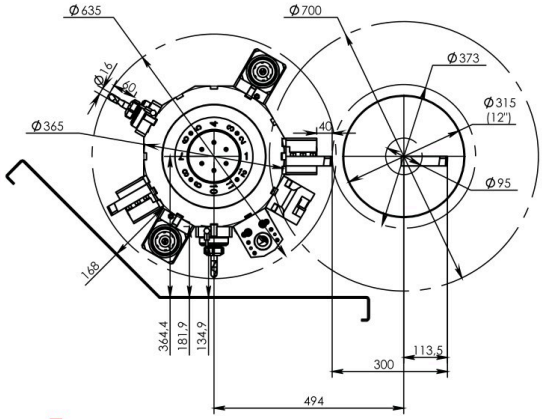
TTC 6-8 C Tool Interference



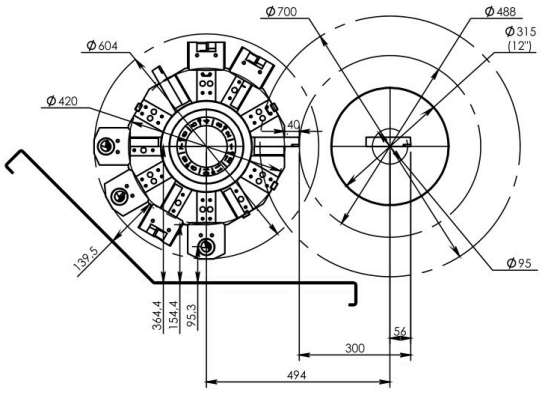
TTH 8-10 C Opt. Tool Interference



TTH 8-10 Std. Tool Interference



TTH 12 C Opt. Tool Interference



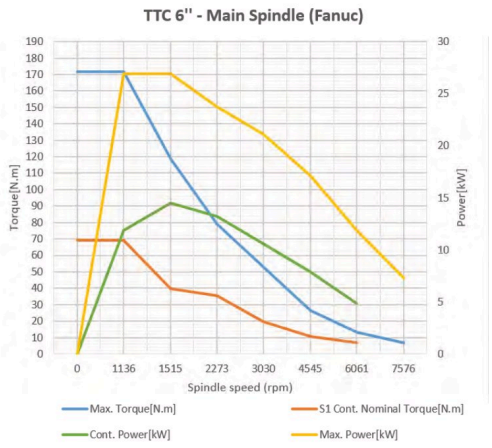
TTH 12 Std. Tool Interference

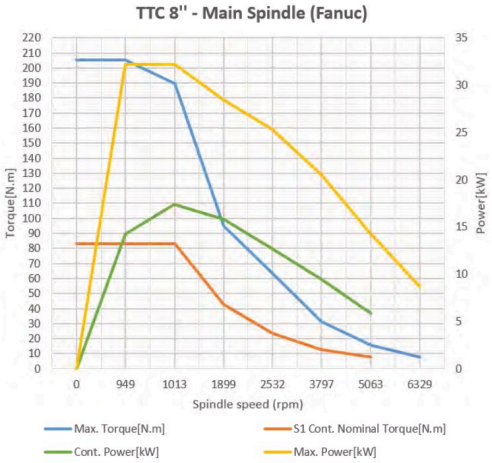
Dimensions on the drawings are in millimetres (1" = 25.4 mm). C Opt. = with the C-axis turret (option); Std. = standard turret.

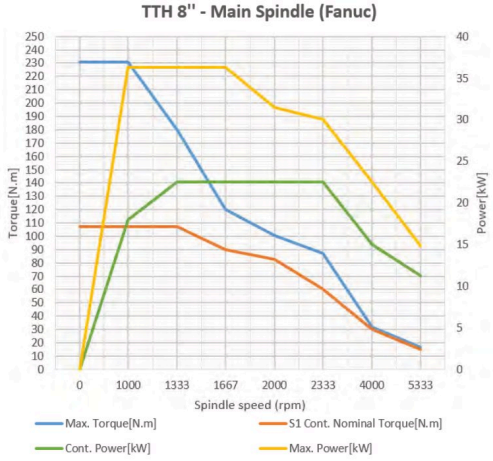
TTC-TTH SERIES · POWER | TORQUE

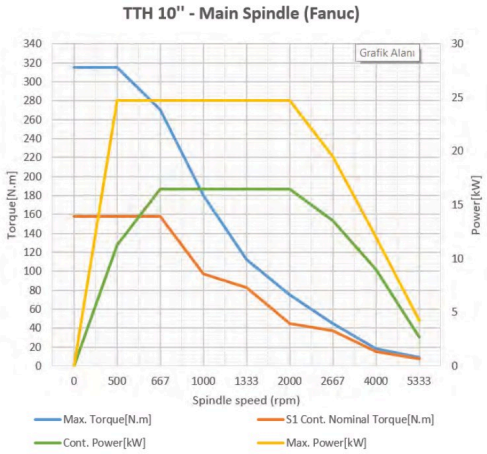
POWER AND TORQUE — FANUC

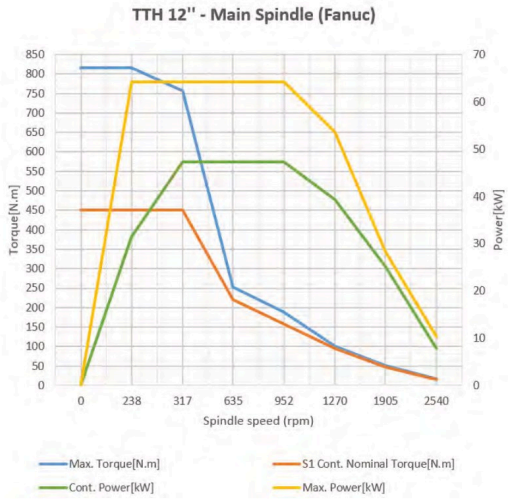
FANUC









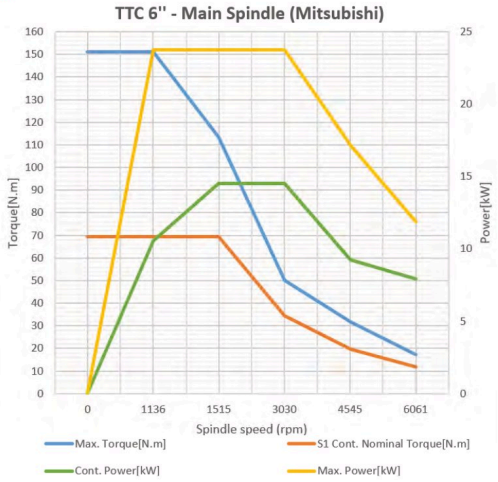


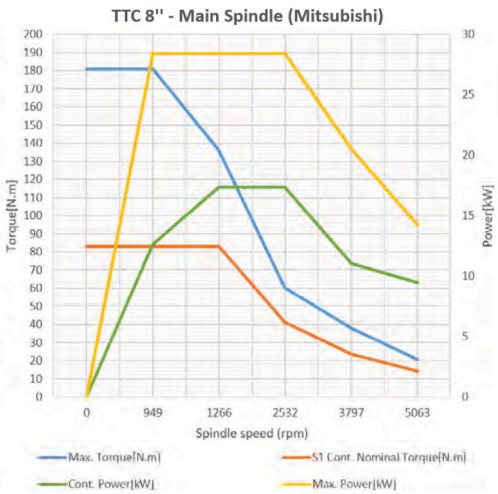
Torque in N·m (1 N·m = 0.738 lbf·ft), power in kW (1 kW = 1.341 HP), spindle speed in rpm. Max. torque, S1 continuous nominal torque, continuous power and max. power of the main spindle motor.

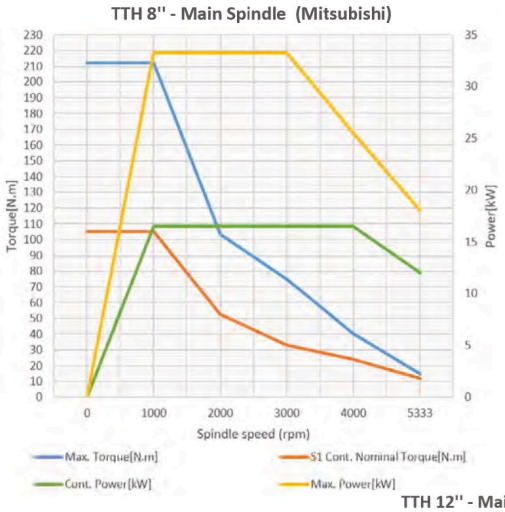
TTC-TTH SERIES · POWER | TORQUE

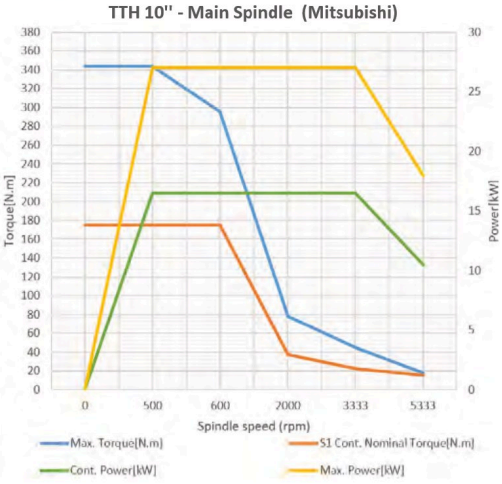
POWER AND TORQUE — MITSUBISHI

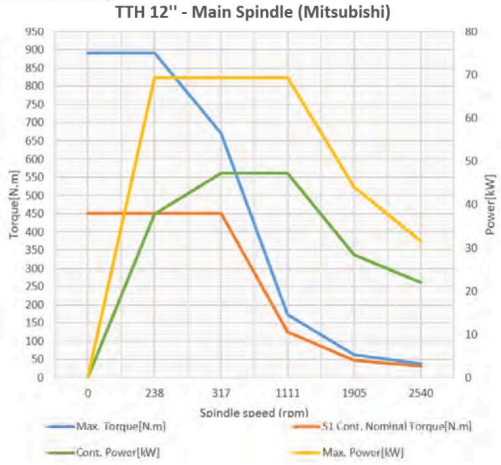
MITSUBISHI









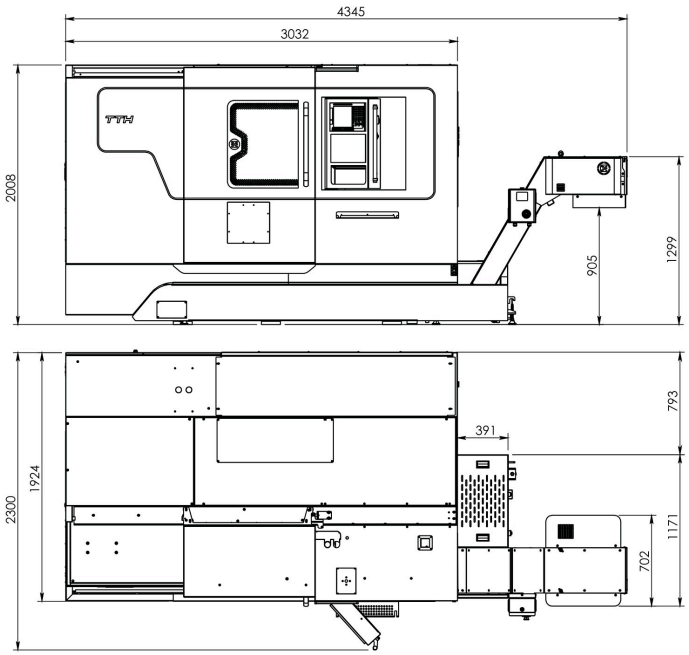


Torque in N·m (1 N·m = 0.738 lbf·ft), power in kW (1 kW = 1.341 HP), spindle speed in rpm. Max. torque, S1 continuous nominal torque, continuous power and max. power of the main spindle motor.

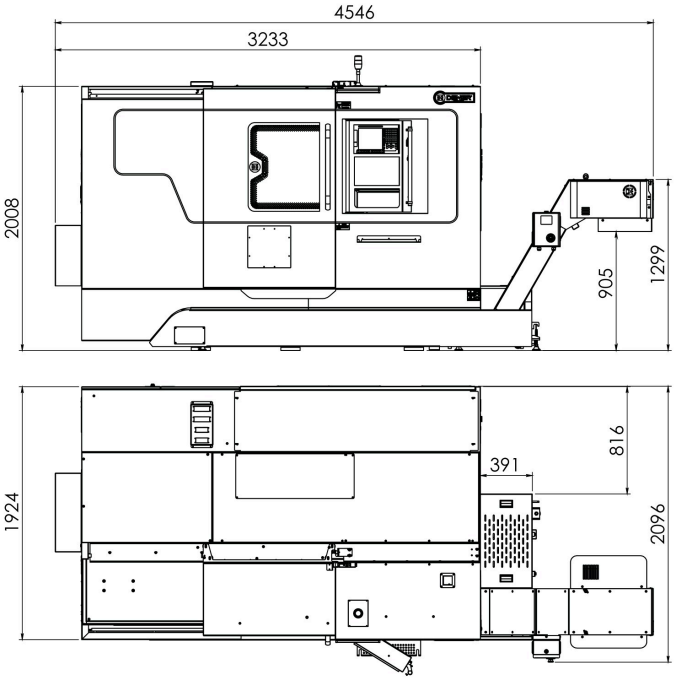
TTC-TTH SERIES · OUTLINE DRAWINGS

TTH AND TTH 12 OUTLINE

TTH



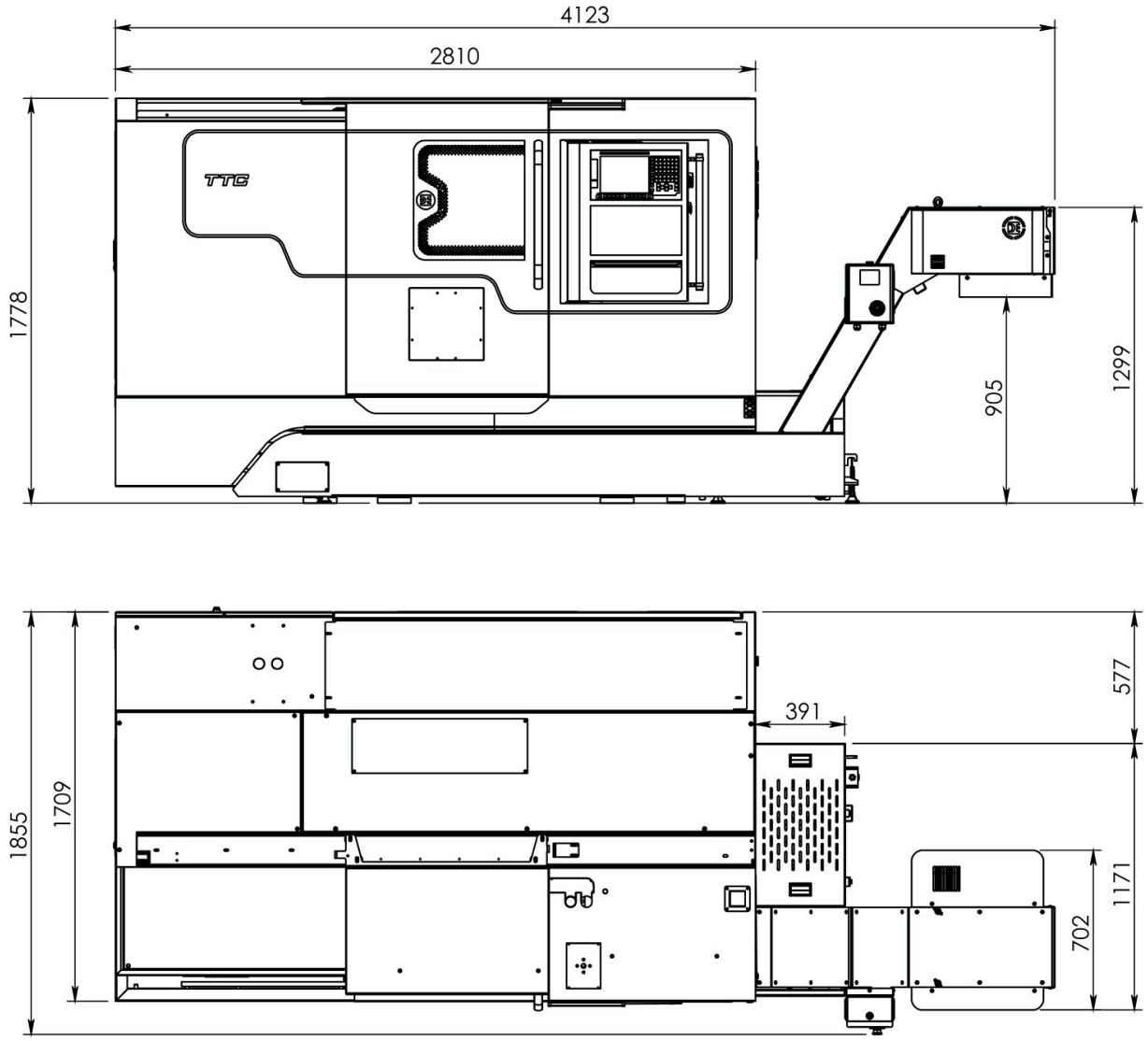
TTH 12



Dimensions on the drawings are in millimetres (1" = 25.4 mm).

TTC-TTH SERIES · OUTLINE DRAWINGS

TTC OUTLINE



Dimensions on the drawings are in millimetres (1" = 25.4 mm).

TTC-TTH SERIES CNC LATHES

TECHNICAL DATA · 1 OF 3

Mitsubishi M80 control as standard, Fanuc 0i-TFP as an option: where the two differ, both values are given.

Model	TTC-6	TTC-8	TTH-8	TTH-10	TTH-12
AXIS MOVEMENTS					
Travel — X axis	7.5″ (190 mm)	7.5″ (190 mm)	11.8″ (300 mm)	11.8″ (300 mm)	11.8″ (300 mm)
Travel — Z axis	17.7″ (450 mm)	17.7″ (450 mm)	29.9″ (760 mm)	29.9″ (760 mm)	29.9″ (760 mm)
AXIS MOTORS					
Feed motor power — X axis Mitsubishi M80 (std.)	3 HP (2.2 kW)	3 HP (2.2 kW)	4 HP (3 kW)	4 HP (3 kW)	4 HP (3 kW)
Feed motor power — X axis Fanuc 0i-TFP (opt.)	2.4 HP (1.8 kW)	2.4 HP (1.8 kW)	4 HP (3 kW)	4 HP (3 kW)	4 HP (3 kW)
Feed motor power — Z axis Mitsubishi M80 (std.)	3 HP (2.2 kW)	3 HP (2.2 kW)	3 HP (2.2 kW)	3 HP (2.2 kW)	3 HP (2.2 kW)
Feed motor power — Z axis Fanuc 0i-TFP (opt.)	2.4 HP (1.8 kW)	2.4 HP (1.8 kW)	2.4 HP (1.8 kW)	2.4 HP (1.8 kW)	2.4 HP (1.8 kW)
Feed motor torque — X axis continuous / holding / max. · Mitsubishi M80 (std.)	8.9 / 23.6 lbf·ft (12 / 32 N·m)	8.9 / 23.6 lbf·ft (12 / 32 N·m)	16.6 / 47.2 lbf·ft (22.5 / 64 N·m)	16.6 / 47.2 lbf·ft (22.5 / 64 N·m)	16.6 / 47.2 lbf·ft (22.5 / 64 N·m)
Feed motor torque — X axis continuous / holding / max. · Fanuc 0i-TFP (opt.)	8.1 / 19.9 lbf·ft (11 / 27 N·m)	8.1 / 19.9 lbf·ft (11 / 27 N·m)	14.8 / 33.2 lbf·ft (20 / 45 N·m)	14.8 / 33.2 lbf·ft (20 / 45 N·m)	14.8 / 33.2 lbf·ft (20 / 45 N·m)
Feed motor torque — Z axis continuous / holding / max. · Mitsubishi M80 (std.)	8.9 / 23.6 lbf·ft (12 / 32 N·m)	8.9 / 23.6 lbf·ft (12 / 32 N·m)	8.9 / 23.6 lbf·ft (12 / 32 N·m)	8.9 / 23.6 lbf·ft (12 / 32 N·m)	8.9 / 23.6 lbf·ft (12 / 32 N·m)
Feed motor torque — Z axis continuous / holding / max. · Fanuc 0i-TFP (opt.)	8.1 / 19.9 lbf·ft (11 / 27 N·m)	8.1 / 19.9 lbf·ft (11 / 27 N·m)	8.1 / 19.9 lbf·ft (11 / 27 N·m)	8.1 / 19.9 lbf·ft (11 / 27 N·m)	8.1 / 19.9 lbf·ft (11 / 27 N·m)
MAIN SPINDLE					
Max. spindle speed	6,000 rpm	4,500 rpm	4,500 rpm	3,500 rpm	2,450 rpm
Main spindle motor power Mitsubishi M80 (std.)	14.8 / 24.1 HP (11 / 18 kW)	14.8 / 24.1 HP (11 / 18 kW)	14.8 / 29.8 HP (11 / 22.2 kW)	14.8 / 24.1 HP (11 / 18 kW)	20.1 / 29.8 HP (15 / 22.2 kW)
Main spindle motor power Fanuc 0i-TFP (opt.)	14.8 / 27.4 HP (11 / 20.4 kW)	14.8 / 27.4 HP (11 / 20.4 kW)	20.1 / 32.5 HP (15 / 24.2 kW)	14.8 / 22.1 HP (11 / 16.5 kW)	20.1 / 27.4 HP (15 / 20.4 kW)
Max. spindle torque Mitsubishi M80 (std.)	111.6 lbf·ft (151.3 N·m)	126.8 lbf·ft (171.9 N·m)	156.4 lbf·ft (212 N·m)	253.6 lbf·ft (343.8 N·m)	656.8 lbf·ft (890.5 N·m)
Max. spindle torque Fanuc 0i-TFP (opt.)	126.6 lbf·ft (171.6 N·m)	143.8 lbf·ft (195 N·m)	170.4 lbf·ft (231 N·m)	232.3 lbf·ft (315 N·m)	601.8 lbf·ft (815.9 N·m)
Max. spindle motor torque 30 min / continuous · Mitsubishi M80 (std.)	38.7 / 84.5 lbf·ft (52.5 / 114.6 N·m)	38.7 / 84.5 lbf·ft (52.5 / 114.6 N·m)	51.6 / 104.2 lbf·ft (70 / 141.3 N·m)	86.1 / 169.1 lbf·ft (116.7 / 229.2 N·m)	105.6 / 208.5 lbf·ft (143.2 / 282.7 N·m)
Max. spindle motor torque 30 min / continuous · Fanuc 0i-TFP (opt.)	38.7 / 95.9 lbf·ft (52.5 / 130 N·m)	38.7 / 95.9 lbf·ft (52.5 / 130 N·m)	52.8 / 113.6 lbf·ft (71.6 / 154 N·m)	77.4 / 154.9 lbf·ft (105 / 210 N·m)	105.5 / 191 lbf·ft (143 / 259 N·m)
Spindle nose	ASA A2-5	ASA A2-6	ASA A2-6	ASA A2-8	ASA A2-11
Spindle bearing diameter (front) inner / outer	3.1 / 4.9″ (80 / 125 mm)	3.9 / 5.9″ (100 / 150 mm)	3.9 / 5.9″ (100 / 150 mm)	5.1 / 7.9″ (130 / 200 mm)	6.3 / 9.4″ (160 / 240 mm)
Spindle bore	2.2″ (56 mm)	2.6″ (66 mm)	2.6″ (66 mm)	3.6″ (92 mm)	4.6″ (118 mm)

TTC-TTH SERIES CNC LATHES

TECHNICAL DATA · 2 OF 3

Model	TTC-6	TTC-8	TTH-8	TTH-10	TTH-12
TURRET					
Number of tool stations	12	12	12	12	12
Tool shank size	1 × 1″ (25 × 25 mm)	1 × 1″ (25 × 25 mm)	1 × 1″ (25 × 25 mm)	1 × 1″ (25 × 25 mm)	1 × 1″ (25 × 25 mm)
Max. boring bar diameter	1.6″ (40 mm)	1.6″ (40 mm)	1.6″ (40 mm)	1.6″ (40 mm)	1.6″ (40 mm)
Type	Slot type	Slot type	Slot type	Slot type	Slot type
Turret indexing time (1 station)	0.20 – 0.34 s	0.20 – 0.34 s	0.20 – 0.34 s	0.20 – 0.34 s	0.20 – 0.34 s
Turret drive type	Servo mechanical	Servo mechanical	Servo mechanical	Servo mechanical	Servo mechanical
CONTROL SYSTEM					
CNC control unit Mitsubishi M80 (std.)	M80 (FCA80H-8A)	M80 (FCA80H-8A)	M80 (FCA80H-8A)	M80 (FCA80H-8A)	M80 (FCA80H-8A)
CNC control unit Fanuc 0i-TFP (opt.)	FANUC 0i-TFP	FANUC 0i-TFP	FANUC 0i-TFP	FANUC 0i-TFP	FANUC 0i-TFP
Screen	10.4″ colour LCD	10.4″ colour LCD	10.4″ colour LCD	10.4″ colour LCD	10.4″ colour LCD
CAPACITY					
Swing	22″ (560 mm)	22″ (560 mm)	27.6″ (700 mm)	27.6″ (700 mm)	27.6″ (700 mm)
Recommended turning diameter	6.7″ (170 mm)	8.3″ (210 mm)	8.3″ (210 mm)	10″ (254 mm)	12.4″ (315 mm)
Max. turning diameter	12″ (305 mm)	12″ (305 mm)	20.8″ (528 mm)	20.8″ (528 mm)	19.2″ (488 mm)
Max. turning length	17″ (433 mm)	16″ (406 mm)	26.8″ (680 mm)	26″ (660 mm)	24.2″ (614 mm)
Chuck size	6″	8″	8″	10″	12″
Bar capacity	1.8″ (45 mm)	2″ (52 mm)	2″ (52 mm)	3″ (75 mm)	3.7″ (95 mm)
Bed angle	45°	45°	45°	45°	45°
Ball screw diameter × pitch and tolerance class — Z axis	C1	C1	C1	C1	C1
Ball screw diameter × pitch and tolerance class — X axis	C3	C3	C3	C3	C3
Guideway size — Z axis	1.18″ (30 mm)	1.18″ (30 mm)	1.77″ (45 mm)	1.77″ (45 mm)	1.77″ (45 mm)
Guideway size — tailstock	1.38″ (35 mm)	1.38″ (35 mm)	1.77″ (45 mm)	1.77″ (45 mm)	1.77″ (45 mm)
Guideway size — X axis	1.18″ (30 mm)	1.18″ (30 mm)	1.77″ (45 mm)	1.77″ (45 mm)	1.77″ (45 mm)
Guideway type [linear]	Roller type	Roller type	Roller type	Roller type	Roller type
AXIS FEEDS					
Rapid traverse — X axis	98 ft/min (30 m/min)	98 ft/min (30 m/min)	98 ft/min (30 m/min)	98 ft/min (30 m/min)	98 ft/min (30 m/min)
Rapid traverse — Z axis	98 ft/min (30 m/min)	98 ft/min (30 m/min)	98 ft/min (30 m/min)	98 ft/min (30 m/min)	98 ft/min (30 m/min)
Cutting feed — X axis	3.3 – 16.4 ft/min (1 – 5 m/min)	3.3 – 16.4 ft/min (1 – 5 m/min)	3.3 – 32.8 ft/min (1 – 10 m/min)	3.3 – 32.8 ft/min (1 – 10 m/min)	3.3 – 32.8 ft/min (1 – 10 m/min)
Cutting feed — Z axis	3.3 – 16.4 ft/min (1 – 5 m/min)	3.3 – 16.4 ft/min (1 – 5 m/min)	3.3 – 32.8 ft/min (1 – 10 m/min)	3.3 – 32.8 ft/min (1 – 10 m/min)	3.3 – 32.8 ft/min (1 – 10 m/min)
ACCURACY					
Positioning accuracy X / Z	0.0003 / 0.00016″ (0.008 / 0.004 mm)	0.0003 / 0.00016″ (0.008 / 0.004 mm)	0.0003 / 0.00016″ (0.008 / 0.004 mm)	0.0003 / 0.00016″ (0.008 / 0.004 mm)	0.0003 / 0.00016″ (0.008 / 0.004 mm)
Repeatability X / Z	0.00016 / 0.00008″ (0.004 / 0.002 mm)	0.00016 / 0.00008″ (0.004 / 0.002 mm)	0.00016 / 0.00008″ (0.004 / 0.002 mm)	0.00016 / 0.00008″ (0.004 / 0.002 mm)	0.00016 / 0.00008″ (0.004 / 0.002 mm)

TTC-TTH SERIES CNC LATHES

TECHNICAL DATA · 3 OF 3

Model	TTC-6	TTC-8	TTH-8	TTH-10	TTH-12
TAILSTOCK					
Tailstock body travel	17.7″ (450 mm)	17.7″ (450 mm)	25.6″ (650 mm)	25.6″ (650 mm)	25.6″ (650 mm)
Quill diameter	Ø2.68″ (Ø68 mm)	Ø2.68″ (Ø68 mm)	Ø2.68″ (Ø68 mm)	Ø2.68″ (Ø68 mm)	Ø2.68″ (Ø68 mm)
Quill inner taper	MK 4	MK 4	MK 4	MK 4	MK 5
POWER AND COOLANT					
Electrical power connection Mitsubishi M80 (std.)	32 kVA	32 kVA	32 kVA	32 kVA	32 kVA
Electrical power connection Fanuc 0i-TFP (opt.)	25 kVA	25 kVA	32 kVA	25 kVA	32 kVA
Coolant pump	1 HP (58 psi) (0.75 kW (4 bar))	1 HP (58 psi) (0.75 kW (4 bar))	1 HP (58 psi) (0.75 kW (4 bar))	1 HP (58 psi) (0.75 kW (4 bar))	1 HP (58 psi) (0.75 kW (4 bar))
Coolant tank capacity	37 US gal (140 L)	37 US gal (140 L)	42 US gal (160 L)	42 US gal (160 L)	42 US gal (160 L)
MACHINE DIMENSIONS					
Length Mitsubishi M80 (std.)	162.1″ (4,118 mm)	162.3″ (4,123 mm)	171.1″ (4,346 mm)	171.1″ (4,346 mm)	179″ (4,546 mm)
Length Fanuc 0i-TFP (opt.)	162.3″ (4,123 mm)	162.3″ (4,123 mm)	171.1″ (4,345 mm)	171.1″ (4,345 mm)	178.9″ (4,545 mm)
Width Mitsubishi M80 (std.)	67.3″ (1,709 mm)	67.3″ (1,709 mm)	81.6″ (2,073 mm)	81.6″ (2,073 mm)	75.7″ (1,924 mm)
Width Fanuc 0i-TFP (opt.)	67.3″ (1,709 mm)	67.3″ (1,709 mm)	75.7″ (1,924 mm)	75.7″ (1,924 mm)	81.6″ (2,073 mm)
Height	70″ (1,778 mm)	70″ (1,778 mm)	79.1″ (2,008 mm)	79.1″ (2,008 mm)	79.1″ (2,008 mm)
Weight (approx.)	7,187 lb (3,260 kg)	7,288 lb (3,306 kg)	11,881 lb (5,389 kg)	11,991 lb (5,439 kg)	12,976 lb (5,886 kg)
C AXIS (OPTION)					
Max. live tool speed	6,000 rpm	6,000 rpm	6,000 rpm	6,000 rpm	5,000 rpm
Live tool motor power continuous / max. · Mitsubishi M80 (std.)	5 / 8.9 HP (3.7 / 6.6 kW)	5 / 8.9 HP (3.7 / 6.6 kW)	5 / 8.9 HP (3.7 / 6.6 kW)	5 / 8.9 HP (3.7 / 6.6 kW)	5 / 8.9 HP (3.7 / 6.6 kW)
Live tool motor power continuous / max. · Fanuc 0i-TFP (opt.)	5 / 11.1 HP (3.7 / 8.3 kW)	5 / 11.1 HP (3.7 / 8.3 kW)	5 / 11.1 HP (3.7 / 8.3 kW)	5 / 11.1 HP (3.7 / 8.3 kW)	5 / 11.1 HP (3.7 / 8.3 kW)
Live tool motor torque continuous / max. · Mitsubishi M80 (std.)	17.4 / 31 lbf·ft (23.6 / 42 N·m)	17.4 / 31 lbf·ft (23.6 / 42 N·m)	17.4 / 31 lbf·ft (23.6 / 42 N·m)	17.4 / 31 lbf·ft (23.6 / 42 N·m)	17.4 / 31 lbf·ft (23.6 / 42 N·m)
Live tool motor torque continuous / max. · Fanuc 0i-TFP (opt.)	13.1 / 38.7 lbf·ft (17.7 / 52.5 N·m)	13.1 / 38.7 lbf·ft (17.7 / 52.5 N·m)	13.1 / 38.7 lbf·ft (17.7 / 52.5 N·m)	13.1 / 38.7 lbf·ft (17.7 / 52.5 N·m)	13.1 / 38.7 lbf·ft (17.7 / 52.5 N·m)
Turret type	BMT55	BMT55	BMT55	BMT55	BMT65
Max. turning diameter	10.8″ (275 mm)	10.8″ (275 mm)	17.2″ (438 mm)	17.2″ (438 mm)	14.7″ (373 mm)
Max. turning length	14.3″ (362 mm)	13.2″ (335 mm)	24.7″ (627 mm)	23.9″ (607 mm)	21.1″ (537 mm)

Specifications from Dener's catalogue. Custom voltage available: 208 V, 240 V, 480 V or 600 V 3-phase. CSA-approved electrical panel.

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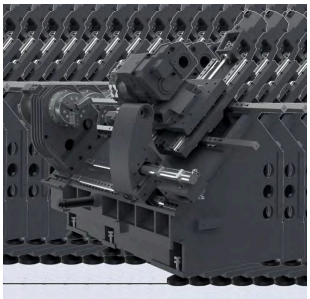
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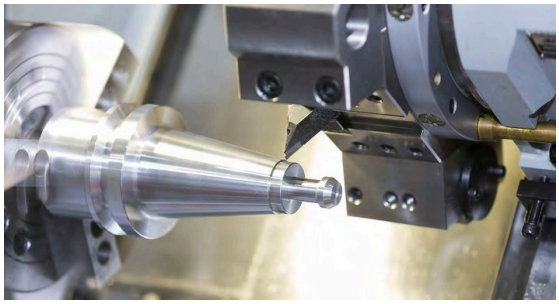
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Specifications subject to change without notice. Imperial values are converted from the manufacturer's metric data and rounded: plate thicknesses in common fractions (1/2, 1/4, 1/8, 1/16). Model numbers are manufacturer designations. Photos may show optional equipment.